

N 1.100

Дано:  
 $j=1; i=5$   
 $|Q| = 2|\Delta U|$   
 $V_1 = 2V_2$   
 $P_1$   
 Найти:  $P_2$

Решение:

$$-2\Delta U = \Delta U - A \Rightarrow -A = 3\Delta U = +\frac{15}{2}JR\Delta T$$

$$C = \frac{\Delta Q}{dT} = \frac{\frac{5}{2}JRdT - \frac{15}{2}JRdT}{dT} = -5R$$

$$n = \frac{-5R - \frac{7}{2}R}{-5R - \frac{5}{2}R} = \frac{-17/2 R}{-15/2 R}$$

$$2^n \cdot P_1 V_1^n = P_2 V_2^n$$

$$\Rightarrow P_2 = P_1 2^{\frac{17}{15}}$$

N 2.6

Дано:  
 $M_{Ar} = 0,04 \frac{кг}{моль}$   
 $M_{He} = 0,004 \frac{кг}{моль}$   
 $\frac{J_A}{J} = 0,01$   
 Найти:  $\frac{\Delta C}{C}$

Решение:

$$C = \sqrt{\frac{fRT}{M}} \Rightarrow \ln C = \ln \sqrt{fRT} - \frac{1}{2} \ln M$$

$$\frac{dC}{C} = -\frac{1}{2} \frac{dM}{M}; \quad dM = M_{Ar+He} - M_{He}$$

$$M = \frac{\sum m}{\sum \nu} = \frac{0,998 \cdot 4 \cdot 10^{-3} + 0,01 \cdot 4 \cdot 10^{-2}}{1} = 0,00436$$

$$\left| \frac{\Delta C}{C} \right| = 0,045$$

Ответ:  $\frac{\Delta C}{C} = 4,5\%$

N 1.35

Дано:

$$\kappa: \text{H}_2 = 2:1$$

$$P_1 = 8 \text{ атм}$$

$$T_1 = 600 \text{ К}$$

$$P_2 = 1 \text{ атм}$$

$$\text{Найти: } T_2$$

$$\begin{cases} P_1 V_1 = \nu R T_1 & P_1 V_1^n = P_2 V_2^n \\ P_2 V_2 = \nu R T_2 \end{cases}$$

$$\bar{C}_V = \frac{\frac{1}{2} \nu \cdot C_{V\text{H}_2} + \frac{2}{2} \nu \cdot C_{V\text{He}}}{\frac{\nu}{2} + \frac{2\nu}{2}} = 2R$$

$$\bar{C}_P = 3R$$

$$\eta = \frac{3}{2} \Rightarrow \left( \frac{V_1}{V_2} \right)^{\frac{3}{2}} = \frac{P_2}{P_1} = \frac{1}{8} \Rightarrow \frac{V_1}{V_2} = \frac{1}{4} \Rightarrow$$

$$\Rightarrow 8 \cdot \frac{1}{4} = \frac{T_1}{T_2} \Rightarrow \underline{T_2 = 300 \text{ К}}$$

$$\text{Ответ: } T_2 = 300 \text{ К}$$